

enforcement is inconsistent and often lacks best practices. Blockchain, on the other hand, transforms the situation by establishing a secure, transparent and tamperproof system for managing and verifying the ownership of digital assets.

An excellent example of this is non-fungible tokens or NFTs. These are digital tokens on a blockchain that certify the ownership of tangible and intangible assets, like a piece of digital art, music, or even a video. Contrary to cryptocurrencies, such as Bitcoin, which are fungible and interchangeable with one another, each NFT is unique and has intrinsic worth. This helps to prove ownership rights over creative works.

Blockchain enables artists to tokenise their artwork. This means that, legally, they are owners of the work and have control over how it is distributed or sold. Each transaction that occurs in this case is recorded in a public ledger such that the full history of ownership is transparent to anyone, yet immutable. This simplifies the job of proving provenance and authenticity. This system reduces the threat posed by art fraud, piracy, and unauthorised reproduction to ensure that creators get paid for their creations.

Smart contracts are another excellent feature of blockchain technology. These self-executing contracts enforce contractual terms and conditions automatically. For instance, when an artist sells her work to someone, it could be set in the contract that she receives a certain percentage of the sale in case the artwork is sold again.

Blockchain's potential for conducting secure elections

Blockchain makes voting more secure, transparent and accessible -- all without compromising privacy. A voting system based on a blockchain allows ballots to be cast and tallied in a distributed and immutable ledger. As soon as a vote is cast, the encrypted entry is added to the blockchain, becoming part of a public record that can never be modified or tampered with. Once the vote has been

cast, it cannot be changed, deleted or duplicated, thereby greatly reducing the chance of fraud at elections.

Blockchain also enhances electoral accessibility. Anyone can vote from anywhere around the world, which is especially useful for expatriates or those with disabilities who cannot vote in the traditional way. In addition, because of the transparency of blockchain, votes can be audited in real time, making the election process more transparent. Furthermore, voting would be open only to eligible voters, thus reducing the chances of impersonation or duplicate voting.

However, while several governments and organisations have begun pilot testing their versions of a blockchain-based voting system, the technology is still in its infancy.

Blockchain in healthcare: Streamlined medical records and drug traceability

The most remarkable impact of blockchain adoption will be felt in hospitals where medical records are kept and maintained. Medical records are often stored in different institutions in disparate forms and hence are not easy to access. Blockchain's decentralised ledger can be used to access information by authorised doctors and the patients themselves. Blockchain technology ensures that such historic data cannot be modified or falsified.

This decentralised mode also enhances patient privacy. Using advanced encryption techniques, sensitive information can be safeguarded from private individuals gaining undue access to patients' private data.

The drug supply chain can benefit enormously from blockchain technology. Counterfeit drugs have long been a challenge in the pharmaceutical industry and have endangered the lives of patients. Adding the flow of a drug through production, distribution, and sales on a ledger that cannot be manipulated offers immutable history about its storage, movement,

and authenticity. So pharmaceutical companies and regulators can easily monitor the entire supply chain for safety standard compliance, and consumers can feel assured that they are buying genuine products.

Blockchain can also safeguard the data integrity of clinical trials. Researchers may use the blockchain to document trial information, making these records available to scrutiny designed to expose manipulation and guarantee trial results are credible.

Blockchain and the Metaverse: Facilitating true ownership in virtual worlds

Central to the Metaverse is the idea of digital ownership, whether these are virtual lands, avatars, NFTs, digital arts, or in-game items. As discussed earlier, NFTs on the blockchain are a way for users to create, buy, sell, and prove ownership of unique digital assets. In this ownership model, the asset, once created, is forever tracked by the blockchain.

On traditional online platforms or games, major stakeholders usually are large companies such as developers of the software. The Metaverse has a decentralised setup, where smart contracts guarantee equitable and transparent management of virtual spaces. This makes users feel in charge of the worlds they inhabit.

Blockchain permits digital identity management in the Metaverse. Just as it gives a user secure and verifiable identity in real life, it allows a distributed digital identity of the user through various virtual spaces. This identity can be augmented against reputation, and digital assets and activities, offering continuity and trust between different platforms and experiences.

Blockchain-backed systems allow micropayments inside virtual spaces to buy, sell, and trade digital goods and services without relying on centralised financial institutions. For the creators and businesspersons in the